Web Programming

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HTML - Structuring the web

Multimedia & embedding

Complete beginners start here!

- Getting started with the web

HTML — Structuring the web

- Introduction to HTML
- Multimedia and embedding
  - Multimedia and embedding
  - Images in HTML
  - Video and audio content
  - From object to iframe — other embedding technologies
  - Adding vector graphics to the web
  - Responsive images
  - Mozilla splash page
- HTML tables

CSS — Styling the web

- CSS first steps
- CSS building blocks
- Styling text
- CSS layout

Multimedia and embedding

We've looked at a lot of text so far in this course, but the web would be really boring only using text. Let's start looking at how to make the web come alive with more interesting content! This module explores how to use HTML to include multimedia in your web pages, including the different ways that images can be included, and how to embed video, audio, and even entire webpages.

In this article

- Prerequisites
- Guides
- Assessments
- See also

Looking to become a front-end web developer?

We have put together a course that includes all the essential information you need to work towards your goal.

Get started

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Prerequisites

Before starting this module, you should have a reasonable understanding of the basics of HTML, as previously covered in Introduction to HTML. If you have not worked through this module (or something similar), work through it first, then come back!

Note: If you are working on a computer/tab
to create your own files, you could try out (
program such as JSBin or Glitch.

https://tinyurl.com/3n52m4r3

Images in HTML

Lecture: Images in HTML
Module: Multimedia & embedding

Introduction

Objective of the lecture

Objective: To learn how to **embed** simple **images** in **HTML**, annotate them with **captions**, and how HTML **images** relate to **CSS background images**.

How do we **put** an **image** on a **webpage**?

The `<img>` element and its `src` attribute

- **Empty element.**
  - ✓Meaning that it has **no text content** or **closing tag**.
- **`<img>`** requires a **minimum** of **one attribute** to be **useful.**
  - ✓`src`
- The `src` **attribute** contains a **path** pointing to the **image** you want to **embed** in the **page.**
  - ✓The **path** can be a **relative** or **absolute URL**.

The `<img>` element and its `src` attribute

- If **your image** is **called** `dinosaur.jpg`, and it is in the **same directory** as your **HTML page**, you could **embed** the image **like so**:

```
1 <img src="dinosaur.jpg">
```

- If the **image** is in an `images` **subdirectory**, which is **inside** the **same directory** as the **HTML page** then you'd **embed** it like this:

```
1 <img src="images/dinosaur.jpg">
```

✓**Google recommends** this **approach** for **SEO/indexing** purposes.

Note

Note: **Search engines** also **read image filenames** and count them **towards SEO**. Therefore, you should give your **image** a **descriptive filename**; **dinosaur.jpg** is better than **img835.png**.

The <img> element and its src attribute

- You could **embed** the **image** using its **absolute URL**:

```
1 <img src="https://www.example.com/images/dinosaur.jpg">
```

- This is **pointless:**
  - ✓it **makes** the **browser** do **more work**, **looking up the IP address** from the **DNS server all over again**, etc.
  - ✓You'll **almost always** keep the **images** for your **website** on the **same server** as your **HTML**.



Warning

Warning: Most images are **copyrighted**. Do **not display** an **image** on your webpage **unless**:

- You **own** the **image**.
- You have **received** explicit, **written permission** from the **image owner**.
- You have ample **proof** that the **image** is, in fact, in the **public domain**.



Warning

Copyright violations are **illegal** and **unethical**. **Never** point your `src` **attribute** at an **image hosted** on **someone else's website** that you **don't have permission** to **link** to. This is called "**hotlinking**". **Stealing someone's bandwidth** is **illegal**. It also **slows down** your **page**, leaving you with **no control over** whether the **image** is **removed** or **replaced** with something **embarrassing**.

Output



html/multimedia-and-embedding/images-in-html/index.html

Note

Note: **Elements** like `<img>` and `<video>` are sometimes referred to as **replaced elements**. This is because the **element's content** and **size** are **defined** by an **external resource** (like an **image** or **video file**), not by the **contents** of the **element itself**.

The `alt` attribute of the `<img>` element

- **Textual description** of the **image.**
- For **use** in **situations** where the **image cannot be seen/displayed** or **takes a long time** to **render** because of a **slow internet connection**.

```
1  <img src="images/dinosaur.jpg" alt="The head and torso of a dinosaur skeleton; it has
2  a large head with long sharp teeth">
```

The `alt` attribute of the `<img>` element

- The **easiest way** to test your **alt text** is to **purposely misspell** your **filename**.
- For **example**, using `dinosooooor.jpg` **instead** of `dinosaur.jpg`



Reasons to use the `alt` attribute

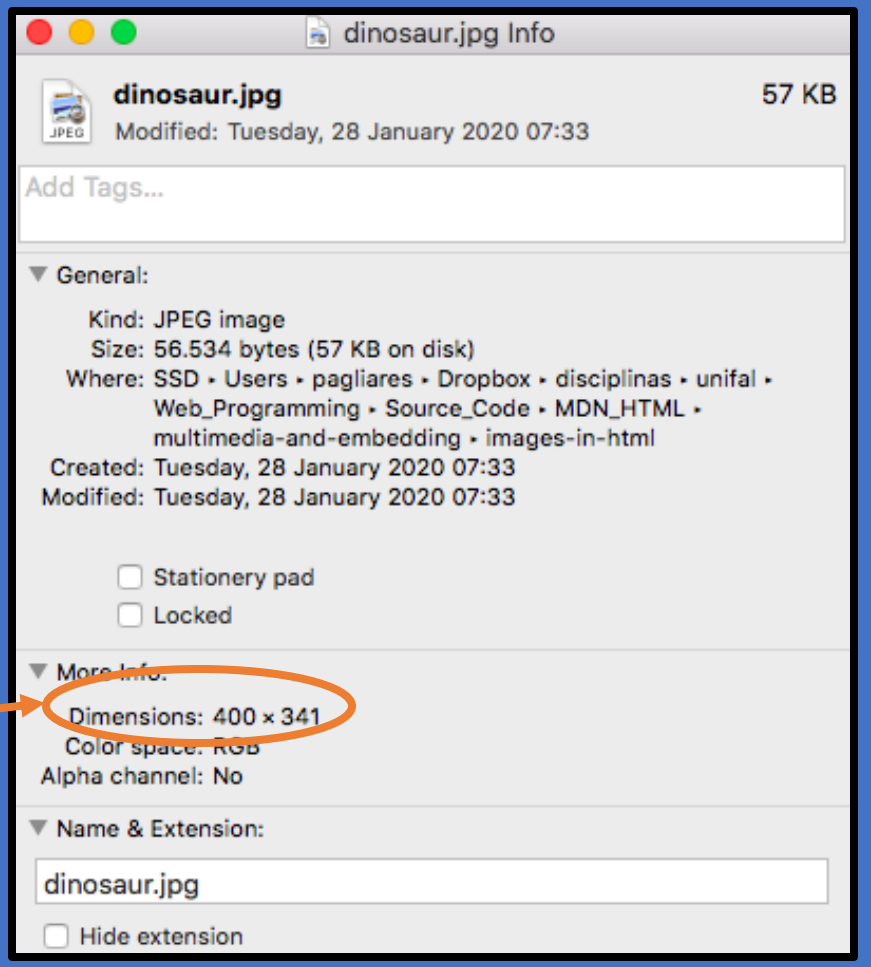
- The **user** is **visually impaired** and is using a **screen reader** to read the **web** out to **them**.
- The **spelling** of the **file** or **path name** might be **wrong**.
- The **browser doesn't support** the **image type**.
- You may want to **provide text** for **search engines** to **utilize**.
- **Users** have **turned off images** to **reduce data transfer volume** and **distractions**.

The **key** is to **deliver** a **usable experience**, even when the **images can't** be **seen**.

`width` and `height` attributes of `<img>` element

- You can **find** your **image's width and height** in **a number of ways**.
  - ✓`Cmd+i` on **Mac OS**.

```
1  <img src="images/dinosaur.jpg" alt="The head and torso of a
2  dinosaur skeleton; it has a large head with long sharp teeth"
3  width="400"
4  height="341">
```

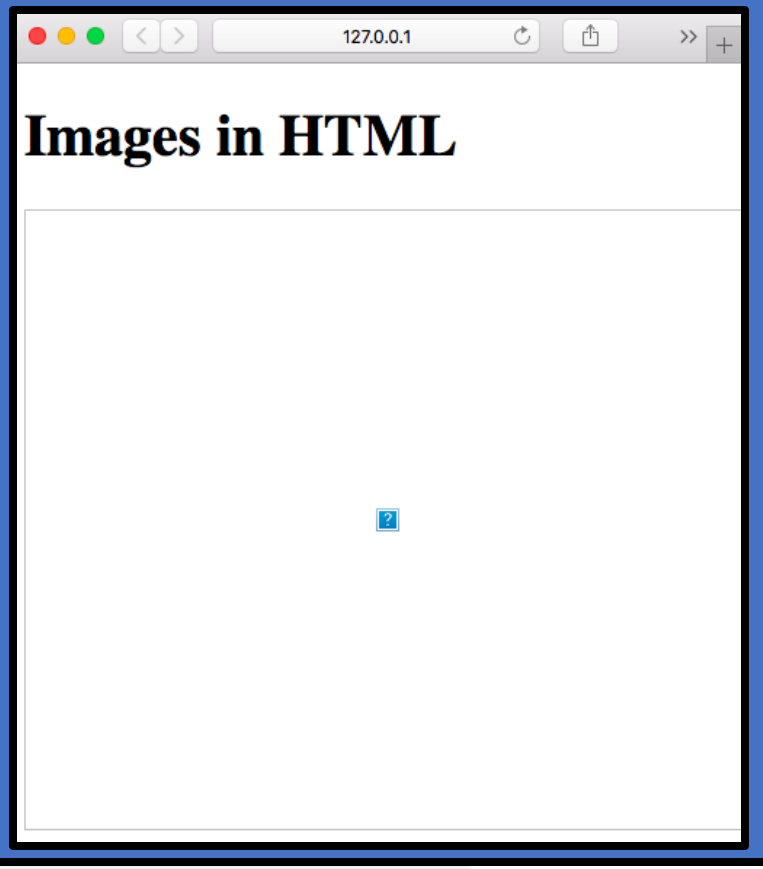


`width` and `height` attributes of `<img>` element

- When the **image** is **not rendered**, we can **notice** that the **web browser leaves** a **space** for the **image** to **appear in**.
  - ✓**`width`** and **`height`** **attributes**.

- At the time of **creation** of these **slides**, **Safari** leaves a **space,** but **doesn't show** the **`alt` attribute**, while **Google Chrome**, **doesn't leave** a **space**, **but shows** the **contents** of the **`alt` attribute**.

SAFARI



CHROME



`width` and `height` attributes of `<img>` element

- You **shouldn't alter** the **size** of your **images** using **HTML attributes**.
  - ✓One **problem** is that the **image may** also **end up looking distorted**, if you **don't maintain** the correct **aspect ratio**.

- **Rationale**:
  - ✓You should **use** an **image editor** to put your **image** at the **correct size before putting** it on your **webpage**.

Note

Note: If you do **need** to **alter** an **image's size**, you should use **CSS** instead.

The `title` attribute of the `<img>` element




```
<img src="images/dinosaur.jpg" alt="The head and torso of a dinosaur skeleton; it has a large
     head with long sharp teeth" width="400"
     height="341"
     title="A T-Rex on display in the Manchester University Museum">
```

Drawbacks of the `title` attribute

- `Title` has a **number** of **accessibility problems**.
  - ✓**Screen reader support** is very **unpredictable**.
  - ✓**Most browsers won't show it unless** you are **hovering** with a **mouse.**
    - **No** access to **keyboard users**.
- It is **better** to **include such supporting information** in the **main article text**, rather than **attached** to the **image**.

Active learning:
Embedding an image

Go to the **link** indicated **below** for the **Active Learning statement**



https://tinyurl.com/2txu783k

Solution



Editable code

Press Esc to move focus away from the code area (Tab inserts a tab character).

```
<img src="https://raw.githubusercontent.com/mdn/learning-area/master/html/multimedia-and-embedding/images-in-
html/dinosaur_small.jpg"
 alt="The head and torso of a dinosaur skeleton; it has a large head with long sharp teeth"
 width="200"
 height="171"
 title="A T-Rex on display in the Manchester University Museum">
```

Annotating images with **figures** and **figure captions**

A possible solution with captions:

- There are a **number** of **ways** that you could **add** a **caption** to go with your **image**.

```
1  <div class="figure">
2    <img src="images/dinosaur.jpg"
3         alt="The head and torso of a
4          dinosaur skeleton; it has a large head with long sharp teeth"
5          width="400" height="341">
6
7    <p>A T-Rex on display in the Manchester University Museum.</p>
8  </div>
```

The **problem** with the `<div>`
Solution for **figure** and **captions**

The problem

- There is a **problem** with the **approach** of the **previous slide**:
  - ✓There is **nothing** that **semantically links** the **image** to its **caption**, which can cause **problems** for **screen readers**.
- When you have **50 images** and **captions**, which **caption** goes with **which image**?

```
1  <div class="figure">
2    <img src="images/dinosaur.jpg"
3          alt="The head and torso of a
4           dinosaur skeleton; it has a large head with long sharp teeth"
5           width="400" height="341">
6
7    <p>A T-Rex on display in the Manchester University Museum.</p>
8  </div>
```

Solving the **problem** with the `<figure>` and `<figcaption>` elements

The <figure> and <figcaption> elements

- A **better solution** for **captions** to use the **HTML5** <figure> and <figcaption> elements.

```
1  <figure>
2    <img src="images/dinosaur.jpg"
3         alt="The head and torso of a dinosaur skeleton; it has a large
4         head with long sharp teeth"
5         width="400"
6         height="341">
7
8     <figcaption>A T-Rex on display in the Manchester University Museum.</figcaption>
9  </figure>
```

Note

Note: From an **accessibility viewpoint**, **captions** and `alt` **text** have **distinct roles**. **Captions benefit even people** who can **see** the **image**, whereas **alt text provides** the **same functionality** as an **absent image**. **Therefore**, **captions** and **alt text shouldn't just say** the **same thing**, **because** they **both appear** when the **image** is **gone**. Try **turning images off** in your **browser** and see how it **looks**.

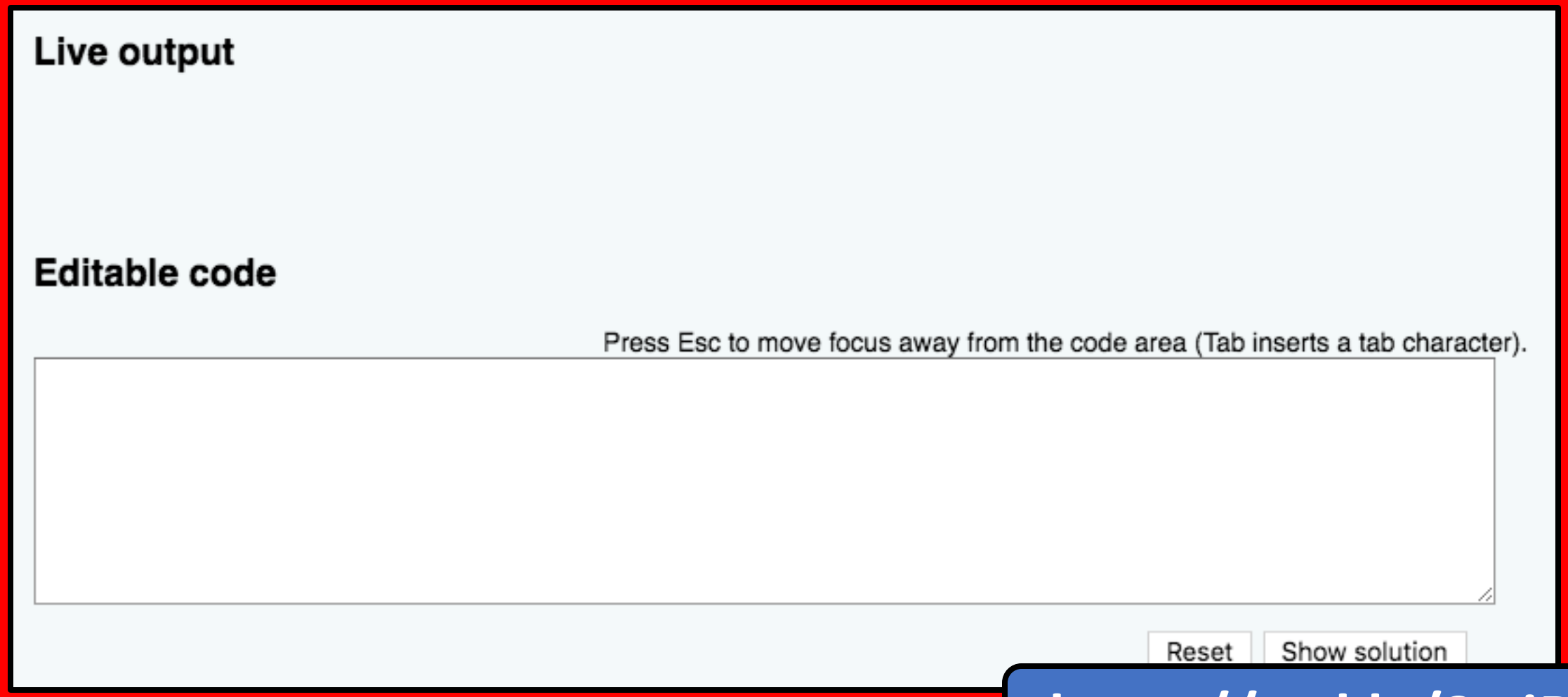
The `<figure>` element

- A **figure doesn't** have to **be** an **image**. It is an **independent unit** of content.

- It can **be**:
  - ✓Several **images**
  - ✓A **code snippet**
  - ✓**Audio**
  - ✓**Video**
  - ✓**Equations**
  - ✓**Table**
  - ✓….

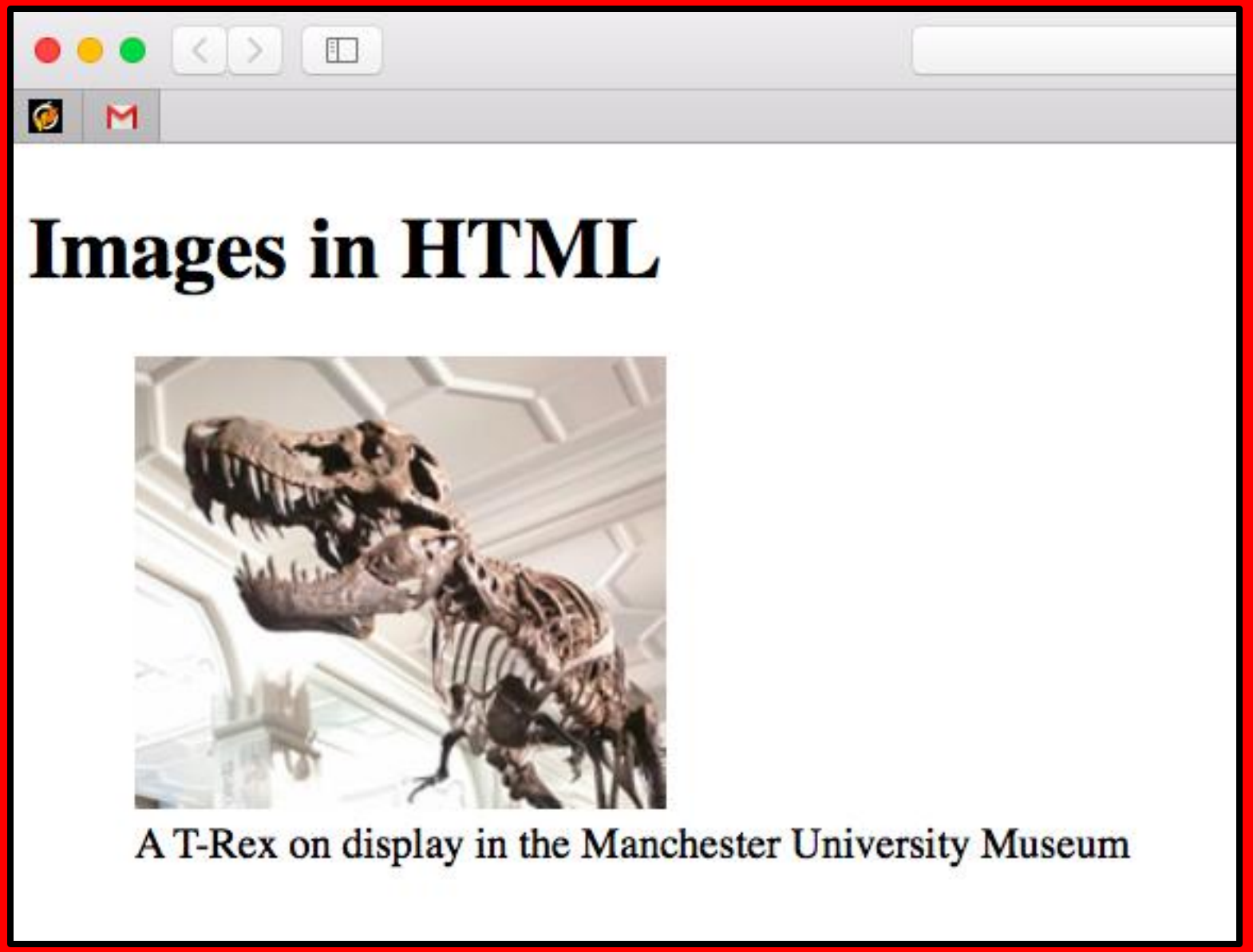
Active learning: Creating a **figure**

Go to the **link** indicated **below** for the **Active Learning statement**



https://mzl.la/2wjRjuj

Solution


Images in HTML
A T-Rex on display in the Manchester University Museum

CSS background images

The CSS `background-image` property

- You can use **CSS** to **embed images** into **webpages**.
- The **CSS** `background-image` **property**, and the other `background-*` **properties**, are **used** to **control background image placement**.
- **Example**, to **place** a **background image** on **every paragraph** on a **page**:

```
1  p {
2    background-image: url("images/dinosaur.jpg");
3  }
```


04-css-background-images
Rodrigo Martins Pagliares
Change View
https://codepen.io/pagliares/pen/eYWzeyR

HTML or **CSS** for **images** in **web documents**?

Why bother with HTML images?

- The resulting **embedded image** with **CSS**, is arguably **easier** to **position** and **control** than **HTML images**.

- **CSS background images** are for **decoration** only.
  - ✓Such **images** have **no semantic meaning** at all.
  - ✓They **can't have** any **text equivalents**, are **invisible to screen readers**, and so on.

- This is where **HTML images shine**!

HTML or CSS for images in web documents?

- **Does the image** has **meaning**, in **terms** of your **content?**
  - ✓ If yes, **use** an **HTML image**.
- Does the **image** is purely **decoration?**
  - ✓**If yes,** use **CSS background images**.

Test your skills

Table of contents

HTML images 1
HTML images 2
HTML images 3
Assessment or further help

Related Topics

Complete beginners start here!

▶ Getting started with the Web

HTML — Structuring the Web

▶ Introduction to HTML

▼ Multimedia and embedding

Multimedia and embedding overview
Images in HTML
Video and audio content

Test your skills: HTML images

The aim of this skill test is to assess whether you've understood our Images in HTML article.

> **Note**: You can try out solutions in the interactive editors below, however it may be helpful to download the code and use an online tool such as CodePen, jsFiddle, or Glitch to work on the tasks.
>
> If you get stuck, then ask us for help — see the Assessment or further help section at the bottom of this page.

HTML images 1

In this task we want you to embed a simple image of some Blueberries into the page. You need to:

- Add the path to the image to an appropriate attribute to embed it on the page. The image is called `blueberries.jpg`, and it is in a folder inside the current folder called `images`.
- Add some alternative text to an appropriate attribute to describe th
  it.
- Give the `<img>` element an appropriate `width` and `height` so
  aspect ratio, and enough space is left on the page to display it. The image's intrinsic size is 615 x 419

https://mzl.la/2vyUr5f

Questions?